PHYS 210 Electricity and Magnetism

Celebration of Knowledge!

Supplementary Practice 1 - CK #2 preparation

Name: _____ Date: _____

Time: _____ Marks: _____ Practice paper - not an official exam

Attempt all questions. Supporting practice; this topic mix is not the announced main-test format.

Write all rough calculations and final answers on separate paper. Show your working.

Original source numbering is retained inside each question. Answer the entire cropped question.

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1. [____] Tutorial 4 - Question 4

4. A long nonconducting cylinder (radius = 6.0 mm) has a nonuniform volume charge density given by αr^2, where $\alpha = 6.2\ \text{mC/m}^5$ and r is the distance from the axis of the cylinder. What is the magnitude of the electric field at a point 2.0 mm from the axis?
   a. 1.4 N/C
   b. 1.6 N/C
   c. 1.8 N/C
   d. 2.0 N/C
   e. 5.4 N/C

2. [____] Homework 5 - Question 23.1

23.1 •• A point charge $q_1 = +2.40\ \mu\text{C}$ is held stationary at the origin. A second point charge $q_2 = -4.30\ \mu\text{C}$ moves from the point $x = 0.150$ m, $y = 0$ to the point $x = 0.250$ m, $y = 0.250$ m. How much work is done by the electric force on q_2?

3. [____] Homework 6 - Question 24.1

24.1 • The plates of a parallel-plate capacitor are 2.50 mm apart, and each carries a charge of magnitude 80.0 nC. The plates are in vacuum. The electric field between the plates has a magnitude of 4.00×10^6 V/m. What is (a) the potential difference between the plates; (b) the area of each plate; (c) the capacitance?

4. [____] Homework 6 - Question 24.3

24.3 • A parallel-plate air capacitor of capacitance 245 pF has a charge of magnitude 0.148 μC on each plate. The plates are 0.328 mm apart. (a) What is the potential difference between the plates? (b) What is the area of each plate? (c) What is the electric-field magnitude between the plates? (d) What is the surface charge density on each plate?